Aimin Li

Postdoctoral Researcher, KTH Royal Institute of Technology
Division of Information Science and Engineering
Stockholm, Sweden

aiminl@kth.se Website Google Scholar LinkedIn

Education

Ph.D., Information and Communication Engineering *2020 – 2025*
Harbin Institute of Technology, China (GPA rank: 1st) **Best Dissertation Award Nomination**

B.S., Electrical Engineering *2016 – 2020*
Harbin Institute of Technology, China (GPA rank: 2nd) **Best Undergraduate Thesis Award**

Exchange Student, University of California, Riverside, USA *Jul – Aug 2017*

Research Experience

Postdoctoral Researcher, KTH Royal Institute of Technology, Sweden *Oct 2026 – Present*
Host: **Prof. Mikael Skoglund** (IEEE Fellow). Swedish Research Council (VR) project.

Postdoctoral Researcher, Middle East Technical University, Türkiye *Oct 2025 – Oct 2026*
Host: **Prof. Elif Uysal** (IEEE Fellow). ERC Advanced Grant GO-SPACE, jointly with DLR.

Visiting Researcher, A*STAR Institute for Infocomm Research, Singapore *Oct 2023 – Oct 2024*
Host: **Prof. Sumei Sun** (IEEE Fellow). Funded by the China Scholarship Council.

Awards and Honors

Research Awards

3rd Prize, IEEE GLOBECOM 4MT Thesis Competition (5th of 15 finalists worldwide) *2024*
IEEE GLOBECOM Student Travel Grant (awarded to the 15 4MT finalists) *2024*
IEEE ITW Student Travel Grant (IEEE Information Theory Workshop) *2026*
Best Dissertation Award Nomination, HIT (sole nominee of the EE Department) *2025*
Best Undergraduate Thesis Award, HIT (one of two in the EE Department) *2020*

Scholarships and Honors

National Scholarship of China (top 0.2% nationwide) *2024*
National Scholarship of China (top 0.2% nationwide) *2020*
Postdoctoral Research Scholarship (TÜBİTAK 2247-A Program) *2026*
Provincial Outstanding Graduate Award, Heilongjiang (top 0.4% in the province) *2025*
Special Merit Doctoral Scholarship, HIT (8 recipients university-wide) *2020*
Outstanding Participant, Tsinghua University EE Summer Camp (one of 8 nationwide) *2019*
Nominee, NSFC Young Students Basic Research Program (sole nominee in EE) *2024*
HIT "Dianzi" Research Student Program (independent PI; sole Ph.D. recipient in EE) *2023*
China Scholarship Council (CSC) Scholarship (visiting research at A*STAR) *2023*

Competitions

First Prize, Chinese College Student Mathematics Competition *2017*
First Prize, Chinese College Student Mathematics Competition *2016*
Provincial Second Prize, China Undergraduate Mathematical Contest in Modeling *2018*

Research Focus

My research lies at the intersection of information theory, stochastic control, and networking. I study how networked systems should acquire, transmit, and act on information when what matters is not only how much information is delivered, but whether it arrives in time to support a decision. My current research focuses on the following themes.

- **Information Timeliness and Age of Information**
  - Optimal sampling, scheduling, and preemption of status updates
  - Queueing with heavy-tailed delays and tail-risk guarantees on timeliness
  - Timeliness in distributed learning and inference systems
- **Goal-Oriented and Semantic Communications**
  - Value of information and sampling for remote decision making
  - Remote estimation of correlated and spatial random processes
  - Joint design of communication and decision-making in distributed systems
- **Non-Terrestrial and Space Networks**
  - Age-aware satellite random access and multi-satellite fusion
  - Store-and-forward and delay-tolerant networking for deep-space links
  - In-orbit computing and inference under intermittent connectivity
- **Finite-Blocklength Information Theory and Coding**
  - Error-probability bounds and encoder design for rateless codes
  - Short-packet coding for low-latency communication

At a Glance

- **Publications**: 24 journal articles (20 published), including IEEE TIT, TCOM, and TWC, and 23 conference papers, including ISIT, ITW, Allerton, and Asilomar.
- **Citations**: 420 citations, h-index 11, i10-index 14 (Google Scholar, November 2026).
- **Mentoring**: 22 students at three institutions; Ph.D. placements at KTH, Caltech, and HKU.

Publications

Aimin Li in bold; underlined authors are students I mentored; [#] co-first authors; [†] corresponding author.

Journal Articles under Review

[J24] H. Bai[#], **Aimin Li**[#], R. Yao, X. Zhao, T. Zhang, X. Zhang, C. Li, L. Gao, and J. Ma, "Multi-Agent Diffusion Planning with Stable Online Reinforcement Fine-Tuning for Cooperative Driving," **submitted to** *IEEE Transactions on Mobile Computing*, 2026.

[J23] S. Chakraborty, **Aimin Li**[†], Y. İnce, S. Baghaee, and E. Uysal, "ASTRA: Asynchronous Age-Aware Satellite Random Access with Correlated Multi-Satellite Fusion," **submitted to** *IEEE Transactions on Wireless Communications*, 2026.

[J22] A. U. Atasayar, **Aimin Li**[†], Ç. Arı, and E. Uysal, "Age-Optimal Sampling and Routing under Intermittent Links and Energy Constraints," **submitted to** *IEEE Transactions on Mobile Computing*, 2026.

[J21] L. Song, D. Zhang, **Aimin Li**, and T. Zhang, "Source PAC Coding for Low-Latency Secret Key Generation in the Short-Blocklength Regime," **submitted to** *IEEE Transactions on Information Theory*, 2025.

Published Journal Articles

[J20] **Aimin Li**, S. Wu, G. C. F. Lee, and S. Sun, "From Freshness to Effectiveness: Goal-Oriented Sampling for Remote Decision Making," *IEEE Transactions on Information Theory*, 2026.

[J19] **Aimin Li**, X. Chen, S. Wu, G. C. F. Lee, and S. Sun, "Unified Upper Bounds on the ML

Decoding Error Probability of Spinal Codes over Fading Channels," *IEEE Transactions on Wireless Communications*, 2026.

[J18] G. Chen, Y. Deng, **Aimin Li**, and S. Wu, "Permanent Connection Guaranteed Resilient LEO Constellation for Remote Area Communications," *IEEE Transactions on Wireless Communications*, 2026.

[J17] H. Wu, S. Wu, **Aimin Li**, S. Meng, and Q. Zhang, "AoI-Aware Joint Sampling-Buffering-Routing Optimization for Autonomous UAV Swarms via a MARL Approach," *IEEE Transactions on Mobile Computing*, 2026.

[J16] X. Chen, **Aimin Li**, Y. Deng, and S. Wu, "Entropy-Driven Sampling for Remote Estimation in Internet of Things Systems," *IEEE Internet of Things Journal*, 2026.

[J15] E. Uysal and **Aimin Li**, "Goal-Oriented Communications for Interplanetary and Non-Terrestrial Networks," *IEEE Communications Magazine*, 2026.

[J14] **Aimin Li**, S. Wu, X. Chen, and S. Sun, "Error Floor of ML-Decoded Spinal Codes in the Finite Blocklength Regime," *IEEE Transactions on Vehicular Technology*, 2025.

[J13] **Aimin Li**, S. Wu, X. Chen, and S. Sun, "Tight Upper Bounds on the BLER of Spinal Codes over the AWGN Channel," *IEEE Transactions on Communications*, 2025.

[J12] D. Zhang, M. Sun, L. Song, S. Jia, **Aimin Li**, T. Zhang, A. Al-Dulaimi, and S. Mumtaz, "Information Freshness and Timeliness Analysis in the Finite Blocklength Regime for Mission-Critical Applications," *IEEE Transactions on Communications*, 2025.

[J11] J. Ning, **Aimin Li**, G. C. F. Lee, S. Sun, and T. Yang, "MARHO: Hybrid Task Offloading in Maritime MEC via Multi-Agent Reinforcement Learning," *IEEE Open Journal of the Communications Society*, 2025.

[J10] **Aimin Li**, S. Wu, S. Meng, R. Lu, S. Sun, and Q. Zhang, "Toward Goal-Oriented Semantic Communications: New Metrics, Framework, and Open Challenges," *IEEE Wireless Communications*, 2024.

[J9] **Aimin Li**, S. Wu, S. Sun, and J. Cao, "Goal-Oriented Tensor: Beyond Age of Information Toward Semantics-Empowered Goal-Oriented Communications," *IEEE Transactions on Communications*, 2024.

[J8] S. Meng, S. Wu, H. Wu, **Aimin Li**, and Q. Zhang, "Effectiveness-Oriented SAGSIN: Unveiling a Unified Metric and a Comprehensive Framework," *IEEE Network*, 2024.

[J7] S. Meng, S. Wu, **Aimin Li**, and Q. Zhang, "Toward Goal-Oriented Semantic Communications: AoII Analysis of Coded Status Update System under FBL Regime," *IEEE Journal on Selected Areas in Information Theory*, 2023.

[J6] **Aimin Li**, S. Wu, J. Jiao, N. Zhang, and Q. Zhang, "Age of Information with Hybrid-ARQ: A Unified Explicit Result," *IEEE Transactions on Communications*, 2022.

[J5] S. Meng, S. Wu, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang, "Analysis and Optimization of the HARQ-Based Spinal Coded Timely Status Update System," *IEEE Transactions on Communications*, 2022.

[J4] S. Meng, S. Wu, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang, "Partial Self-Concatenation Structure and Performance Analysis of Spinal Codes over Rayleigh Fading Channel," *IEEE Transactions on Vehicular Technology*, 2022.

[J3] S. Wu, Z. Deng, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang, "Minimizing Age-of-Information in HARQ-CC Aided NOMA Systems," *IEEE Transactions on Wireless Communications*, 2022.

[J2] X. Lv, S. Wu, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang, "A Weighted Graph-Based Handover Strategy for Aeronautical Traffic in LEO SatCom Networks," *IEEE Networking Letters*, 2022.

[J1] **Aimin Li**, S. Wu, J. Jiao, N. Zhang, and Q. Zhang, "Spinal Codes over Fading Channel: Error Probability Analysis and Encoding Structure Improvement," *IEEE Transactions on Wireless*

Communications, 2021.

Conference Proceedings

[C23] **Aimin Li** and E. Uysal, "Idle Space: Symbiotic Throughput in AoI-Optimal Status Update Systems," **submitted to** *IEEE Conference on Computer Communications (INFOCOM)*, 2027.

[C22] **Aimin Li**, Y. İnce, and E. Uysal, "Taming the Heavy Tail: Age-Optimal Preemption," *IEEE International Symposium on Information Theory (ISIT)*, 2026.

[C21] S. Chakraborty[#], **Aimin Li**[#], Y. İnce, S. Baghaee, and E. Uysal, "ASTRA: Asynchronous Age-Aware Satellite Random Access via Mean-Field Control," *Asilomar Conference on Signals, Systems, and Computers*, 2026.

[C20] A. N. Pehlivanoğlu[#], **Aimin Li**[#], E. Uysal, S. Ciftci, and M. Ozdem, "In-Orbit Intelligence or Ground Offloading? Inference Freshness under Intermittent Satellite Connectivity," *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*, 2026.

[C19] H. Bai[#], **Aimin Li**[#], R. Yao, X. Zhao, T. Zhang, X. Zhang, L. Gao, and J. Ma, "SCORP: Scene-Consistent Multi-Agent Diffusion Planning with Stable Online Reinforcement Post-Training for Cooperative Driving," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[C18] B. Karasakal[#], **Aimin Li**[#], and E. Uysal, "Age of Information in Time-Varying Multi-Priority Queues," *IEEE Information Theory Workshop (ITW)*, 2026.

[C17] A. U. Atasayar, **Aimin Li**, and E. Uysal, "Worse Routes, Fresher Information: Preemption Redefines Which Route is Best," *ACM MobiHoc Workshops*, 2026.

[C16] Y. Yang, D. Zhang, **Aimin Li**, Y. Shen, A. A. Frolov, and B. Gu, "On the Multi-Weight Shift-Sum Iterative Decoding of Short BCH Codes," *IEEE Global Communications Conference (GLOBECOM)*, 2026.

[C15] A. Lei, D. Zhang, **Aimin Li**, and E. Khorov, "Capacity Analysis of Weibull Fading Channels for Satellite-Ground Integrated Communications," *IEEE Vehicular Technology Conference (VTC2026-Spring)*, 2026.

[C14] S. Baghaee, **Aimin Li**, S. Balasubramanian, and E. Uysal, "Goal-Oriented Bundle Management and Flow Control for Deep-Space Communications via Reinforcement Learning," *International Astronautical Congress (IAC)*, 2026.

[C13] **Aimin Li** and E. Uysal, "Optimal Sampling and Scheduling for Remote Fusion Estimation of Correlated Wiener Processes," *Allerton Conference on Communication, Control, and Computing*, 2025.

[C12] A. U. Atasayar, **Aimin Li**, Ç. Arı, and E. Uysal, "Fresh Data Delivery: Joint Sampling and Routing for Minimizing the Age of Information," *ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc), Poster*, 2025. **Acceptance rate: 23%.**

[C11] K. Liu, S. Wu, **Aimin Li**, and Q. Zhang, "Joint Transmission and Control in a Goal-Oriented NOMA Network," *IEEE International Conference on Communications (ICC)*, 2025.

[C10] **Aimin Li**, S. Wu, G. C. F. Lee, X. Chen, and S. Sun, "Sampling to Achieve the Goal: An Age-Aware Remote Markov Decision Process," *IEEE Information Theory Workshop (ITW)*, 2024.

[C9] **Aimin Li**, S. Wu, and S. Sun, "Goal-Oriented Tensor: Beyond AoI Towards Semantics-Empowered Goal-Oriented Communications," *IEEE Wireless Communications and Networking Conference (WCNC)*, 2024.

[C8] X. Chen, **Aimin Li**, and S. Wu, "Optimal Sampling for Uncertainty-of-Information Minimization in a Remote Monitoring System," *IEEE Information Theory Workshop (ITW)*, 2024.

[C7] M. Sun, D. Zhang, S. Jia, and **Aimin Li**, "Packet Management of AoI in the Finite Block-

Length Regime," *IEEE International Conference on Communication Technology (ICCT)*, 2024.

[C6] X. Chen[#], **Aimin Li**[#], and S. Wu, "Tight Upper Bounds on the Error Probability of Spinal Codes over Fading Channels," *IEEE International Symposium on Information Theory (ISIT)*, 2023.

[C5] K. Liu, **Aimin Li**, and S. Wu, "Deep Reinforcement Learning-Assisted Age-Optimal Transmission Policy for HARQ-Aided NOMA Networks," *IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2023.

[C4] **Aimin Li**, S. Wu, Y. Deng, J. Jiao, N. Zhang, and Q. Zhang, "Analyzing Age Performance of Hybrid-ARQ: A Unified Explicit Result," *IEEE Global Communications Conference (GLOBECOM)*, 2022.

[C3] J. Zhang, S. Wu, **Aimin Li**, J. Jiao, and Q. Zhang, "Analysis of GEO Satellite Relay Coded Systems," *IEEE Vehicular Technology Conference (VTC2022-Fall)*, 2022.

[C2] S. Meng, S. Wu, **Aimin Li**, J. Jiao, N. Zhang, and Q. Zhang, "Analysis and Optimization of Spinal Codes over the BSC: From the AoI Perspective," *IEEE International Conference on Communications Workshops (ICC Workshops)*, 2021.

[C1] **Aimin Li**, S. Wu, Y. Wang, J. Jiao, and Q. Zhang, "Spinal Codes over BSC: Error Probability Analysis and the Puncturing Design," *IEEE Vehicular Technology Conference (VTC2020-Spring)*, 2020.

Theses

[T2] **Aimin Li**, "Key Technologies for Timeliness-Enhanced Space-Air Communications in Perception-Decision Closed-Loop Systems," *Ph.D. Thesis, Harbin Institute of Technology*, 2025. **Best Dissertation Award Nomination.**

[T1] **Aimin Li**, "Performance Analysis and Encoding Algorithm Design of Spinal Codes in the Finite Blocklength Regime," *B.S. Thesis, Harbin Institute of Technology*, 2020. **Best Undergraduate Thesis Award.**

Research Grants and Projects

As Principal Investigator

- *HIT "Dianzi" Research Student Program*, independent PI research project, Harbin Institute of Technology, RMB 20,000, 2023.

As Researcher

- *Swedish Research Council (VR) project on joint communication and decision-making for distributed systems*, PI: Prof. Mikael Skoglund, KTH, 2026 – present.
- *ERC Advanced Grant GO-SPACE*, PI: Prof. Elif Uysal, METU, jointly with DLR, 2025 – 2026.

Teaching and Mentoring

Teaching

Teaching Assistant, *Foundations of Information Theory* (graduate), HIT — *Fall 2022*
Teaching Assistant, *Communication Theory* (undergraduate), HIT — *Fall 2020*

Research Mentoring

I have initiated and led research collaborations with **22** junior students across three institutions, shaping their research questions, models, and solution frameworks. My mentees have continued to Ph.D. programs at **KTH**, **Caltech**, and **HKU**. Parentheses give my authorship role on each paper.

Middle East Technical University (METU), 2025 – 2026

- **Adem Utku Atasayar** (M.S., 2025–26): co-supervised thesis research on sampling and routing, leading to MobiHoc [C12, C17] (**co-author**) and a TMC submission [J22] (**co-author**); now a Ph.D. student at METU.
- **Burak Karasakal** (B.S., 2025–26): mentored research on multi-priority queues, leading to ITW [C18] (**co-first author**); recipient of the **IEEE ITW Student Travel Grant (2026)** and admitted to the Ph.D. program at **Caltech**.
- **Yiğit İnce** (M.S., 2025–26): mentored research on preemption and random access, leading to ISIT [C22] (**first author**), Asilomar [C21] (**co-first author**), and a TWC submission [J23] (**co-author**).
- **Ayşe Nur Pehlivanoğlu** (B.S./M.S., 2025–26): mentored research on satellite inference freshness, leading to WiSEE [C20] (**co-first author**).
- **Sayam Chakraborty** (B.S., 2026): mentored thesis research on satellite random access, leading to Asilomar [C21] (**co-first author**) and a TWC submission [J23] (**co-author**); now at ISRO.
- **S. Balasubramanian** (B.S., 2026): mentored thesis research on deep-space communications, leading to IAC [C14] (**co-author**); admitted to the M.S. program at **NUS**.
- **Çağrı Arı** (Ph.D.): collaborated on sampling and routing research, including MobiHoc [C12] (**co-author**); now at Aselsan.
- **Süleyman Çıtır** (M.S.): mentored through research collaboration.
- **Selay Nisa Şahin** (B.S. intern, 2026): supervised a research internship.
- **Yusuf** (B.S. intern, 2026): supervised a research internship.
- **Aral** (B.S. intern, 2026): supervised a research internship.

Harbin Institute of Technology, Shenzhen (HIT), 2020 – 2026

- **Siqi Meng** (B.S./Ph.D., 2020–24): co-supervised undergraduate thesis and doctoral research on timely and goal-oriented communications, leading to WCM, TCOM, TVT, JSAIT, and IEEE Network (**first author**: [J10]; **co-author**: [J5, J4, J7, J8]).
- **Xiaomeng Chen** (B.S./M.S., 2023–26): co-supervised undergraduate and master's theses on coding and remote estimation, leading to TWC, IoTJ, ISIT, and ITW (**first author**: [J19]; **co-author**: [J16, C8]; **co-first author**: [C6]); admitted to the Ph.D. program at **KTH**.
- **Haoxu Wu** (Ph.D., 2025–26): co-supervised research on AoI-aware UAV swarm scheduling, leading to TMC [J17] (**co-author**).
- **Haojie Bai** (Ph.D.): collaborated on research on cooperative driving, leading to IROS [C19] (**co-first author**) and a TMC submission [J24] (**co-first author**); now at Huawei.
- **Kunpeng Liu** (B.S./M.S., 2024–26): co-supervised undergraduate and master's theses on NOMA networks, leading to ICC and INFOCOM Workshops [C11, C5] (**co-author**); now at Huawei.
- **Jiaming Zhang** (B.S./M.S., 2022–25): mentored undergraduate thesis and master's research on satellite relay coded systems, leading to VTC [C3] (**co-author**).
- **Xin Jin** (B.S., 2022): co-supervised the undergraduate thesis.

Zhengzhou University (ZZU), 2025 – 2026

- **Aiwei Lei** (B.S., 2026): mentored research on satellite-ground fading channels, leading to VTC [C15] (**co-author**); admitted to the Ph.D. program at **HKU**.
- **Yinglei Yang** (M.S., 2026): mentored research on short BCH code decoding, leading to GLOBECOM [C16] (**co-author**).
- **Mingxiao Sun** (M.S., 2025): mentored research on finite-blocklength information freshness, leading to TCOM and ICCT [J12, C7] (**co-author**).
- **Lulu Song** (Ph.D., 2026): mentored research on short-blocklength secret key generation, leading to a TIT submission [J21] (**co-author**).

SERVICE

Editorial Service

- **Leading Guest Editor**, *Electronics* (MDPI) Special Issue "From Bits to Decisions", 2026
- **Guest Editor**, *Entropy* (MDPI) Special Issue "Information Freshness and Semantic Communications", 2026 – 2027

Conference Organization

- **Workshop Co-Chair**, IEEE ICC Workshop on *World Models for 6G-Native Networked Robotics*, Washington, D.C., USA *Jun. 2027*
- **Session Chair**, Age of Information, IEEE ITW, Shenzhen, China *Nov. 2024*
- **Session Chair**, Distributed Learning, IEEE GLOBECOM, Cape Town, South Africa *Dec. 2024*
- **TPC Member**, ICFEEIE, Huangshan, China *Jun. 2024*

Journal Reviewer

- IEEE Transactions on Information Theory
- IEEE Journal on Selected Areas in Communications
- IEEE/ACM Transactions on Networking
- IEEE Transactions on Mobile Computing
- IEEE Transactions on Communications
- IEEE Transactions on Wireless Communications
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Vehicular Technology
- IEEE Transactions on Cognitive Communications and Networking
- IEEE Internet of Things Journal
- IEEE Communications Letters
- IEEE Wireless Communications Letters
- *Engineering*
- *Scientific Reports*
- Elsevier Knowledge-Based Systems
- Elsevier Computer Networks
- Elsevier Computer Communications
- Elsevier Ad Hoc Networks
- AEÜ International Journal of Electronics and Communications
- MDPI *Electronics*

Conference Reviewer

- IEEE GLOBECOM, Macau, China *Dec. 2026*
- IEEE ISIT, Guangzhou, China *Jun. 2026*
- IEEE ITW, Tempe, AZ, USA *Nov. 2026*
- IEEE WCNC, Kuala Lumpur, Malaysia *Apr. 2026*
- IEEE VTS APWCS, Seoul, Korea *Aug. 2026*
- IEEE GLOBECOM, Taipei, Taiwan *Dec. 2025*
- IEEE GLOBECOM, Cape Town, South Africa *Dec. 2024*
- IEEE INFOCOM, Vancouver, Canada *May 2024*
- IEEE WCNC, Dubai, UAE *Apr. 2024*
- IEEE ICMLCN, Stockholm, Sweden *May 2024*
- ICFEEIE, Huangshan, China *Jun. 2024*
- IEEE GLOBECOM, Kuala Lumpur, Malaysia *Dec. 2023*
- IEEE ISIT, Taipei, Taiwan *Jun. 2023*
- IEEE/CIC ICCC, Dalian, China *Aug. 2023*
- CSAE, Virtual Event, China *Oct. 2023*
- IEEE GLOBECOM, Rio de Janeiro, Brazil *Dec. 2022*
- IEEE WCNC, Austin, TX, USA *Apr. 2022*
- IEEE VTC *2022*

Volunteer Service

- IEEE VTC2024-Spring, Singapore *Jun. 2024*
- IEEE 6G Summit, Singapore *2024*
- IEEE 6G Summit, Singapore *2023*

Talks

Taming the Heavy Tail: Age-Optimal Preemption

- IEEE ISIT 2026, Guangzhou, China *Jul 2026*
- **Invited Talk**, Sun Yat-sen University, Shenzhen, China *Jul 2026*

4MT Thesis Presentation, IEEE GLOBECOM 2024, Cape Town, South Africa
3rd Prize, IEEE GLOBECOM 4MT Thesis Competition *Dec 2024*

Sampling to Achieve the Goal: An Age-Aware Remote Markov Decision Process
IEEE ITW 2024, Shenzhen, China *Nov 2024*

Goal-Oriented Tensor: Beyond AoI Towards Semantics-Empowered Goal-Oriented Communications
IEEE WCNC 2024, Dubai, UAE *Apr 2024*

Sampling and Scheduling for Goal-Oriented Communications
Invited Talk, HIT and Peng Cheng Laboratory, Shenzhen, China *May 2023*

Analyzing Age Performance of Hybrid-ARQ: A Unified Explicit Result
IEEE GLOBECOM 2022, Rio de Janeiro, Brazil, *Online* *Dec 2022*

Spinal Codes over BSC: Error Probability Analysis and the Puncturing Design
IEEE VTC2020-Spring, Antwerp, Belgium, *Online* *May 2020*